2.26 Differentiation: Inverse Hyperbolic Functions

Worked Solutions

1)

Let $y = \operatorname{arsinh} x$. Then $\sinh y = x$, so

$$\begin{aligned}\frac{d}{dx}(\sinh y) &= \frac{d}{dx}(x),\\ \cosh y\frac{dy}{dx} &= 1,\\ \frac{dy}{dx} &= \frac{1}{\cosh y} = \frac{1}{\sqrt{\sinh^2 y + 1}} = \frac{1}{\sqrt{x^2+1}},\end{aligned}$$

as required.

2)

Let $y = \operatorname{arcosh} x$. Then $\cosh y = x$, so

$$\begin{aligned}\frac{d}{dx}(\cosh y) &= \frac{d}{dx}(x),\\ \sinh y\frac{dy}{dx} &= 1,\\ \frac{dy}{dx} &= \frac{1}{\sinh y} = \frac{1}{\sqrt{\cosh^2 y - 1}} = \frac{1}{\sqrt{x^2-1}},\end{aligned}$$

as required.

3)

Let $y = \operatorname{artanh} x$. Then $\tanh y = x$, so

$$\begin{aligned}\frac{d}{dx}(\tanh y) &= \frac{d}{dx}(x),\\ \operatorname{sech}^2 y\frac{dy}{dx} &= 1,\\ \frac{dy}{dx} &= \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1-\tanh^2 y} = \frac{1}{1-x^2},\end{aligned}$$

as required.

4 a)

$$y = \operatorname{arsinh} 3x \quad \Rightarrow \quad \frac{dy}{dx} = 3 \times \frac{1}{\sqrt{(3x)^2+1}} = \frac{3}{\sqrt{9x^2+1}}.$$

4 b)

$$y = \operatorname{arcosh} 2x \quad \Rightarrow \quad \frac{dy}{dx} = 2 \times \frac{1}{\sqrt{(2x)^2-1}} = \frac{2}{\sqrt{4x^2-1}}.$$

4 c)

$$y = \operatorname{artanh} 5x \quad \Rightarrow \quad \frac{dy}{dx} = 5 \times \frac{1}{1-(5x)^2} = \frac{5}{1-25x^2}.$$

4 d)

$$\begin{aligned} y &= \operatorname{arsinh} \frac{x}{2}, \\ \frac{dy}{dx} &= \frac{1}{2} \times \frac{1}{\sqrt{(x/2)^2+1}} = \frac{1}{2\sqrt{1+x^2/4}} \\ &= \frac{1}{2\sqrt{(x^2+4)/4}} = \frac{1}{\sqrt{x^2+4}}. \end{aligned}$$

4 e)

$$\begin{aligned} y &= \operatorname{arcosh} \frac{x}{4}, \\ \frac{dy}{dx} &= \frac{1}{4} \times \frac{1}{\sqrt{(x/4)^2-1}} = \frac{1}{4\sqrt{x^2/16-1}} \\ &= \frac{1}{4\sqrt{(x^2-16)/16}} = \frac{1}{\sqrt{x^2-16}}. \end{aligned}$$

4 f)

$$\begin{aligned} y &= \operatorname{artanh}\left(-\frac{x}{6}\right), \\ \frac{dy}{dx} &= -\frac{1}{6} \times \frac{1}{1-(-x/6)^2} = -\frac{1}{6(1-x^2/36)} \\ &= -\frac{6}{36-x^2}. \end{aligned}$$

5 a)

$$\begin{aligned} y &= 2\operatorname{arsinh} 2x - 2\operatorname{arcosh} 4x, \\ \frac{dy}{dx} &= 2 \times 2 \times \frac{1}{\sqrt{(2x)^2+1}} - 2 \times 4 \times \frac{1}{\sqrt{(4x)^2-1}} \\ &= \frac{4}{\sqrt{4x^2+1}} - \frac{8}{\sqrt{16x^2-1}}. \end{aligned}$$

5 b)

$$y = \frac{1}{3}\operatorname{artanh} 3x - \frac{2}{3}\operatorname{artanh} 6x,$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1}{3}\times 3\times\frac{1}{1-(3x)^2} - \frac{2}{3}\times 6\times\frac{1}{1-(6x)^2}\\ &= \frac{1}{1-9x^2} - \frac{4}{1-36x^2}\\ &= \frac{(1-36x^2)-4(1-9x^2)}{(1-9x^2)(1-36x^2)}\\ &= -\frac{3}{(1-9x^2)(1-36x^2)}.\end{aligned}$$

6)

$$\begin{aligned}y &= 4\operatorname{arsinh} x,\\ y\big|_{x=2} &= 4\ln(2+\sqrt{2^2+1}) = 4\ln(2+\sqrt{5}),\\ \frac{dy}{dx} &= \frac{4}{\sqrt{x^2+1}},\\ \left.\frac{dy}{dx}\right|_{x=2} &= \frac{4}{\sqrt{5}} = \frac{4\sqrt{5}}{5}.\end{aligned}$$

So the tangent is

$$y - 4\ln(2+\sqrt{5}) = \frac{4\sqrt{5}}{5}(x-2),$$

and hence $y = \frac{4\sqrt{5}}{5}x + 4\ln(2+\sqrt{5}) - \frac{8\sqrt{5}}{5}$.

7)

$$\begin{aligned}y &= -6\operatorname{arcosh}\frac{x}{2},\\ y\big|_{x=4} &= -6\ln\left(2+\sqrt{2^2-1}\right) = -6\ln(2+\sqrt{3}),\\ \frac{dy}{dx} &= -6\times\frac{1}{2}\times\frac{1}{\sqrt{(x/2)^2-1}} = -\frac{6}{\sqrt{x^2-4}},\\ \left.\frac{dy}{dx}\right|_{x=4} &= -\frac{6}{\sqrt{12}} = -\sqrt{3}.\end{aligned}$$

So the tangent is $y + 6\ln(2+\sqrt{3}) = -\sqrt{3}(x-4)$, and hence $y = -\sqrt{3}x + 4\sqrt{3} - 6\ln(2+\sqrt{3})$.

8)

$$\begin{aligned}y &= 10\operatorname{artanh}\frac{x}{5},\\ y\big|_{x=3} &= 10\times\frac{1}{2}\ln\left(\frac{1+3/5}{1-3/5}\right) = 5\ln 4,\\ \frac{dy}{dx} &= 10\times\frac{1}{5}\times\frac{1}{1-(x/5)^2} = \frac{50}{25-x^2},\\ \left.\frac{dy}{dx}\right|_{x=3} &= \frac{50}{16} = \frac{25}{8}.\end{aligned}$$

So the tangent is $y - 5\ln 4 = \frac{25}{8}(x-3)$, and hence $y = \frac{25}{8}x + 5\ln 4 - \frac{75}{8}$.

9)

$$\begin{aligned} y &= \operatorname{arsinh} 4x - \operatorname{arcosh} 2x, \\ \frac{dy}{dx} &= \frac{4}{\sqrt{16x^2+1}} - \frac{2}{\sqrt{4x^2-1}} \\ &= 4(16x^2+1)^{-1/2} - 2(4x^2-1)^{-1/2}, \\ \frac{d^2y}{dx^2} &= -64x(16x^2+1)^{-3/2} + 8x(4x^2-1)^{-3/2}. \end{aligned}$$

At $x = 2$,

$$\begin{aligned} \left.\frac{d^2y}{dx^2}\right|_{x=2} &= -128(65)^{-3/2} + 16(15)^{-3/2} \\ &= -\frac{128\sqrt{65}}{4225} + \frac{16\sqrt{15}}{225} = 0.0311\ldots > 0. \end{aligned}$$

As $\left.\frac{d^2y}{dx^2}\right|_{x=2} > 0$, the curve is convex (concave upwards) at $x = 2$.

10)

$$\begin{aligned} y &= \sqrt{x^2+1} + \operatorname{arsinh} x, \\ \frac{dy}{dx} &= \frac{x}{\sqrt{x^2+1}} + \frac{1}{\sqrt{x^2+1}} = \frac{x+1}{\sqrt{x^2+1}}. \end{aligned}$$

Stationary points exist when $\frac{dy}{dx} = 0$, so $x + 1 = 0$ and $x = -1$. Then

$$y\big|_{x=-1} = \sqrt{2} + \ln(-1+\sqrt{2}).$$

So the stationary point is $\left(-1, \sqrt{2} + \ln(-1+\sqrt{2})\right)$.

11)

$$\begin{aligned} y &= 12 \operatorname{arcosh} x - 3x, \\ \frac{dy}{dx} &= \frac{12}{\sqrt{x^2-1}} - 3. \end{aligned}$$

A stationary point exists when

$$\begin{aligned} \frac{12}{\sqrt{x^2-1}} - 3 &= 0 \\ \frac{12}{\sqrt{x^2-1}} &= 3 \\ 4 &= \sqrt{x^2-1} \\ x &= \sqrt{17}, \end{aligned}$$

since $\operatorname{arcosh} x$ is defined only for $x \geq 1$. At this point

$$\begin{aligned} y &= 12 \operatorname{arcosh}(\sqrt{17}) - 3\sqrt{17} \\ &= 12 \ln(4+\sqrt{17}) - 3\sqrt{17}. \end{aligned}$$

So the range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \leq 12\ln(4+\sqrt{17}) - 3\sqrt{17}\}$.

12 a)

$A(3,0)$.

12 b)

$$\begin{aligned} y &= 2\sqrt{x^2-9} - 12\operatorname{arcosh}\frac{x}{3}, \\ \frac{dy}{dx} &= \frac{2x}{\sqrt{x^2-9}} - \frac{4}{\sqrt{(x/3)^2-1}} \\ &= \frac{2x}{\sqrt{x^2-9}} - \frac{12}{\sqrt{x^2-9}} = \frac{2x-12}{\sqrt{x^2-9}}. \end{aligned}$$

Stationary points exist when $2x - 12 = 0$, so $x = 6$. Then

$$\begin{aligned} y\big|_{x=6} &= 2\sqrt{6^2-9} - 12\operatorname{arcosh} 2 \\ &= 6\sqrt{3} - 12\ln(2+\sqrt{3}). \end{aligned}$$

So the stationary point is $\left(6, 6\sqrt{3} - 12\ln(2+\sqrt{3})\right)$ and the range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \geq 6\sqrt{3} - 12\ln(2+\sqrt{3})\}$.

13 a)

$$\begin{aligned} y &= 4\operatorname{artanh} x + \ln(x+1) - 6x, \\ \frac{dy}{dx} &= \frac{4}{1-x^2} + \frac{1}{x+1} - 6 \\ &= \frac{4+1-x-6(1-x^2)}{1-x^2} = \frac{6x^2-x-1}{1-x^2}. \end{aligned}$$

Stationary points exist when $\frac{dy}{dx} = 0$:

$$\begin{aligned} 6x^2 - x - 1 &= 0, \\ (3x+1)(2x-1) &= 0, \\ x = -\frac{1}{3} \quad &\text{or} \quad x = \frac{1}{2}. \end{aligned}$$

$$\begin{aligned} y\big|_{x=-1/3} &= 4\operatorname{artanh}\left(-\frac{1}{3}\right) + \ln\frac{2}{3} + 2 \\ &= 2\ln\frac{1}{2} + \ln\frac{2}{3} + 2 = 2 - \ln 6, \\ y\big|_{x=1/2} &= 4\operatorname{artanh}\frac{1}{2} + \ln\frac{3}{2} - 3 \\ &= 2\ln 3 + \ln\frac{3}{2} - 3 = \ln\frac{27}{2} - 3. \end{aligned}$$

So the stationary points are $\left(-\frac{1}{3}, 2 - \ln 6\right)$ and $\left(\frac{1}{2}, \ln\frac{27}{2} - 3\right)$.

13 b)

$$\frac{d^2y}{dx^2} = \frac{(1-x^2)(12x-1) - (6x^2-x-1)(-2x)}{(1-x^2)^2}$$

$$= \frac{-x^2 + 10x - 1}{(1 - x^2)^2}.$$

Points of inflection exist when $\frac{d^2y}{dx^2} = 0$:

$$\begin{aligned} -x^2 + 10x - 1 &= 0, \\ x^2 - 10x + 1 &= 0, \\ x &= 5 \pm 2\sqrt{6}. \end{aligned}$$

$5 + 2\sqrt{6} = 9.89$ (3 SF) and $5 - 2\sqrt{6} = 0.101$ (3 SF). The domain of $\operatorname{artanh} x$ is $\{x : x \in \mathbb{R}, -1 < x < 1\}$, so $x \neq 5 + 2\sqrt{6}$. Checking either side of $x = 5 - 2\sqrt{6}$:

$$\left.\frac{d^2y}{dx^2}\right|_{x=0.1} = -0.0102\ldots < 0, \qquad \left.\frac{d^2y}{dx^2}\right|_{x=0.2} = 1.04\ldots > 0.$$

Therefore there is a point of inflection for the curve at $x = 5 - 2\sqrt{6}$.

14 a)

$$\begin{aligned} y &= \operatorname{arsinh}(\sqrt{x}) = \operatorname{arsinh}(x^{1/2}), \\ \frac{dy}{dx} &= \frac{1}{2}x^{-1/2} \times \frac{1}{\sqrt{(x^{1/2})^2 + 1}} = \frac{1}{2\sqrt{x}\sqrt{x+1}} = \frac{1}{2\sqrt{x(x+1)}}. \end{aligned}$$

14 b)

$$\begin{aligned} y &= \operatorname{arcosh}\left(\frac{1}{\sqrt{x}}\right) = \operatorname{arcosh}(x^{-1/2}), \\ \frac{dy}{dx} &= -\frac{1}{2}x^{-3/2} \times \frac{1}{\sqrt{(x^{-1/2})^2 - 1}} \\ &= -\frac{1}{2x^{3/2}\sqrt{x^{-1} - 1}} = -\frac{1}{2\sqrt{x^2(1-x)}}. \end{aligned}$$

14 c)

$$\begin{aligned} y &= \operatorname{artanh}(\sqrt[3]{x}) = \operatorname{artanh}(x^{1/3}), \\ \frac{dy}{dx} &= \frac{1}{3}x^{-2/3} \times \frac{1}{1 - (x^{1/3})^2} \\ &= \frac{1}{3x^{2/3}(1 - x^{2/3})}. \end{aligned}$$

15 a)

$$\frac{d}{dx}\left(\operatorname{arsinh}\frac{1}{x}\right) = \frac{d}{dx}(\operatorname{arsinh} x^{-1})$$

$$= -x^{-2} \times \frac{1}{\sqrt{(x^{-1})^2 + 1}}$$
$$= -\frac{1}{x^2} \times \frac{1}{\sqrt{x^{-2} + 1}}$$
$$= -\frac{1}{x^2} \times \frac{x}{\sqrt{x^2 + 1}} = -\frac{1}{x\sqrt{x^2 + 1}},$$

as required, since $x > 0$.

15 b)

$$y = \sqrt{x^2 + 1} - \operatorname{arsinh} \frac{1}{x},$$
$$\frac{dy}{dx} = \frac{x}{\sqrt{x^2 + 1}} + \frac{1}{x\sqrt{x^2 + 1}}$$
$$= \frac{x^2 + 1}{x\sqrt{x^2 + 1}}.$$

For all $x > 0$, $x^2 + 1 > 1$, so there are no real values of x for which $\frac{x^2+1}{x\sqrt{x^2+1}} = 0$. Hence the curve has no stationary points.

16 a)

$$\frac{d}{dx}\left(\operatorname{artanh} \frac{2x}{\sqrt{4x^2 + 1}}\right) = \frac{\sqrt{4x^2 + 1} \times 2 - 2x \times 8x \times \frac{1}{2}(4x^2 + 1)^{-1/2}}{4x^2 + 1}$$
$$\times \frac{1}{1 - \left(\frac{2x}{\sqrt{4x^2+1}}\right)^2}$$
$$= \frac{2}{(4x^2 + 1)^{3/2}} \times (4x^2 + 1)$$
$$= \frac{2}{\sqrt{4x^2 + 1}},$$

as required.

16 b)

$$y = \sqrt{4x^2 + 1} - \operatorname{artanh} \frac{2x}{\sqrt{4x^2 + 1}},$$
$$\frac{dy}{dx} = \frac{4x}{\sqrt{4x^2 + 1}} - \frac{2}{\sqrt{4x^2 + 1}} = \frac{4x - 2}{\sqrt{4x^2 + 1}}.$$

Stationary points exist when $4x - 2 = 0$, so $x = \frac{1}{2}$. Then

$$y|_{x=1/2} = \sqrt{2} - \operatorname{artanh}\left(\frac{\sqrt{2}}{2}\right)$$
$$= \sqrt{2} - \frac{1}{2}\ln\left(\frac{1 + \sqrt{2}/2}{1 - \sqrt{2}/2}\right)$$
$$= \sqrt{2} - \frac{1}{2}\ln(3 + 2\sqrt{2}).$$

So there is a single stationary point at $\left(\frac{1}{2}, \sqrt{2} - \frac{1}{2}\ln(3 + 2\sqrt{2})\right)$.

17)

$$f(x) = \operatorname{arcosh} 2x + \operatorname{arcosh} \frac{2}{x},$$
$$f'(x) = \frac{2}{\sqrt{4x^2-1}} - \frac{2}{x\sqrt{4-x^2}}.$$

Stationary points exist when $f'(x) = 0$:

$$\frac{1}{\sqrt{4x^2-1}} = \frac{1}{x\sqrt{4-x^2}},$$
$$x\sqrt{4-x^2} = \sqrt{4x^2-1},$$
$$x^2(4-x^2) = 4x^2 - 1,$$
$$x^4 = 1.$$

As $x > 0$, $x = 1$. Then

$$f(1) = \operatorname{arcosh} 2 + \operatorname{arcosh} 2 = 2 \operatorname{arcosh} 2$$
$$= 2\ln(2+\sqrt{3}) = \ln(7+4\sqrt{3}).$$

So there is a stationary point at $(1, \ln(7+4\sqrt{3}))$.

18 a)

$$y = \operatorname{arsinh}(\cosh x),$$
$$\frac{dy}{dx} = \sinh x \times \frac{1}{\sqrt{\cosh^2 x + 1}} = \frac{\sinh x}{\sqrt{\cosh^2 x + 1}}.$$

18 b)

$$y = \operatorname{arcosh}(\tanh x),$$
$$\frac{dy}{dx} = \operatorname{sech}^2 x \times \frac{1}{\sqrt{\tanh^2 x - 1}} = \frac{\operatorname{sech}^2 x}{\sqrt{\tanh^2 x - 1}}.$$

18 c)

$$y = \operatorname{artanh}(\sinh x),$$
$$\frac{dy}{dx} = \cosh x \times \frac{1}{1 - \sinh^2 x} = \frac{\cosh x}{1 - \sinh^2 x}.$$

18 d)

$$y = \operatorname{arsinh}(\tanh x),$$
$$\frac{dy}{dx} = \operatorname{sech}^2 x \times \frac{1}{\sqrt{\tanh^2 x + 1}} = \frac{\operatorname{sech}^2 x}{\sqrt{\tanh^2 x + 1}}.$$

18 e)

$$\begin{aligned} y &= \operatorname{arcosh}(\sinh x), \\ \frac{dy}{dx} &= \cosh x \times \frac{1}{\sqrt{\sinh^2 x - 1}} = \frac{\cosh x}{\sqrt{\sinh^2 x - 1}}. \end{aligned}$$

18 f)

$$\begin{aligned} y &= \operatorname{artanh}(\cosh x), \\ \frac{dy}{dx} &= \sinh x \times \frac{1}{1 - \cosh^2 x} = \frac{\sinh x}{1 - \cosh^2 x}. \end{aligned}$$

19 a)

$$\begin{aligned} y &= x^2 \operatorname{arsinh} x, \\ \frac{dy}{dx} &= x^2 \times \frac{1}{\sqrt{x^2 + 1}} + 2x \operatorname{arsinh} x \\ &= \frac{x^2}{\sqrt{x^2 + 1}} + 2x \operatorname{arsinh} x. \end{aligned}$$

19 b)

$$\begin{aligned} y &= \sinh x \operatorname{arsinh} x, \\ \frac{dy}{dx} &= \sinh x \times \frac{1}{\sqrt{x^2 + 1}} + \cosh x \operatorname{arsinh} x \\ &= \frac{\sinh x}{\sqrt{x^2 + 1}} + \cosh x \operatorname{arsinh} x. \end{aligned}$$

19 c)

$$\begin{aligned} y &= \cosh x \operatorname{arcosh} x, \\ \frac{dy}{dx} &= \cosh x \times \frac{1}{\sqrt{x^2 - 1}} + \sinh x \operatorname{arcosh} x \\ &= \frac{\cosh x}{\sqrt{x^2 - 1}} + \sinh x \operatorname{arcosh} x. \end{aligned}$$

19 d)

$$\begin{aligned} y &= \operatorname{arsinh} x \operatorname{arcosh} x, \\ \frac{dy}{dx} &= \operatorname{arsinh} x \times \frac{1}{\sqrt{x^2 - 1}} + \frac{1}{\sqrt{x^2 + 1}} \operatorname{arcosh} x \\ &= \frac{\operatorname{arsinh} x}{\sqrt{x^2 - 1}} + \frac{\operatorname{arcosh} x}{\sqrt{x^2 + 1}}. \end{aligned}$$

19 e)

$$\begin{aligned} y &= \operatorname{artanh} 2x \operatorname{artanh} 4x, \\ \frac{dy}{dx} &= \operatorname{artanh} 2x \times \frac{4}{1-16x^2} + \frac{2}{1-4x^2} \operatorname{artanh} 4x \\ &= \frac{4 \operatorname{artanh} 2x}{1-16x^2} + \frac{2 \operatorname{artanh} 4x}{1-4x^2}. \end{aligned}$$

20 a)

$$\begin{aligned} y &= \frac{x}{\operatorname{arsinh} x}, \\ \frac{dy}{dx} &= \frac{\operatorname{arsinh} x \times 1 - x \times \frac{1}{\sqrt{x^2+1}}}{(\operatorname{arsinh} x)^2} \\ &= \frac{\operatorname{arsinh} x - \frac{x}{\sqrt{x^2+1}}}{(\operatorname{arsinh} x)^2}. \end{aligned}$$

20 b)

$$\begin{aligned} y &= \frac{\operatorname{arsinh} x}{\operatorname{arcosh} x}, \\ \frac{dy}{dx} &= \frac{\operatorname{arcosh} x \times \frac{1}{\sqrt{x^2+1}} - \operatorname{arsinh} x \times \frac{1}{\sqrt{x^2-1}}}{(\operatorname{arcosh} x)^2} \\ &= \frac{\frac{\operatorname{arcosh} x}{\sqrt{x^2+1}} - \frac{\operatorname{arsinh} x}{\sqrt{x^2-1}}}{(\operatorname{arcosh} x)^2}. \end{aligned}$$

20 c)

$$\begin{aligned} y &= \frac{\operatorname{arcosh} 3x}{\operatorname{arsinh} 6x}, \\ \frac{dy}{dx} &= \frac{\operatorname{arsinh} 6x \times \frac{3}{\sqrt{9x^2-1}} - \operatorname{arcosh} 3x \times \frac{6}{\sqrt{36x^2+1}}}{(\operatorname{arsinh} 6x)^2} \\ &= \frac{\frac{3 \operatorname{arsinh} 6x}{\sqrt{9x^2-1}} - \frac{6 \operatorname{arcosh} 3x}{\sqrt{36x^2+1}}}{(\operatorname{arsinh} 6x)^2}. \end{aligned}$$

20 d)

$$\begin{aligned} y &= \frac{\operatorname{arsinh} 4x}{\operatorname{artanh} 8x}, \\ \frac{dy}{dx} &= \frac{\operatorname{artanh} 8x \times \frac{4}{\sqrt{16x^2+1}} - \operatorname{arsinh} 4x \times \frac{8}{1-64x^2}}{(\operatorname{artanh} 8x)^2} \\ &= \frac{\frac{4 \operatorname{artanh} 8x}{\sqrt{16x^2+1}} - \frac{8 \operatorname{arsinh} 4x}{1-64x^2}}{(\operatorname{artanh} 8x)^2}. \end{aligned}$$

21)

$$\begin{aligned} y &= \sinh(2\operatorname{arcosh} x), \\ \frac{dy}{dx} &= \frac{2}{\sqrt{x^2-1}}\cosh(2\operatorname{arcosh} x), \\ \frac{d^2y}{dx^2} &= \frac{4\sqrt{x^2-1}\sinh(2\operatorname{arcosh} x) - 2x\cosh(2\operatorname{arcosh} x)}{(x^2-1)^{3/2}}. \end{aligned}$$

Now

$$\begin{aligned} (x^2-1)\frac{d^2y}{dx^2} + x\frac{dy}{dx} &= \frac{4\sqrt{x^2-1}\sinh(2\operatorname{arcosh} x) - 2x\cosh(2\operatorname{arcosh} x)}{\sqrt{x^2-1}} \\ &\quad + \frac{2x\cosh(2\operatorname{arcosh} x)}{\sqrt{x^2-1}} \\ &= 4\sinh(2\operatorname{arcosh} x) = 4y. \end{aligned}$$

22)

$$\begin{aligned} y &= (\operatorname{arsinh} 4x)^2, \\ \frac{dy}{dx} &= \frac{8\operatorname{arsinh} 4x}{\sqrt{16x^2+1}}, \\ \frac{d^2y}{dx^2} &= \frac{32\sqrt{16x^2+1} - 128x\operatorname{arsinh} 4x}{(16x^2+1)^{3/2}}. \end{aligned}$$

Therefore

$$\begin{aligned} (16x^2+1)\frac{d^2y}{dx^2} + 16x\frac{dy}{dx} &= \frac{32\sqrt{16x^2+1} - 128x\operatorname{arsinh} 4x}{\sqrt{16x^2+1}} + \frac{128x\operatorname{arsinh} 4x}{\sqrt{16x^2+1}} \\ &= 32. \end{aligned}$$

23)

Let $y = \operatorname{arcosech} x$, so $\operatorname{cosech} y = x$. Then

$$\begin{aligned} \frac{d}{dx}(\operatorname{cosech} y) &= \frac{d}{dx}(x), \\ -\coth y \operatorname{cosech} y \frac{dy}{dx} &= 1, \\ \frac{dy}{dx} &= -\frac{1}{\coth y \operatorname{cosech} y} = -\frac{1}{\sqrt{\operatorname{cosech}^2 y + 1}\ \operatorname{cosech} y} \\ &= -\frac{1}{x\sqrt{x^2+1}}, \end{aligned}$$

as required.

24)

Let $y = \operatorname{arcsech} x$, so $\operatorname{sech} y = x$. Then

$$\frac{d}{dx}(\operatorname{sech} y) = \frac{d}{dx}(x),$$

$$
\begin{aligned}
-\tanh y \operatorname{sech} y \frac{dy}{dx} &= 1, \\
\frac{dy}{dx} &= -\frac{1}{\tanh y \operatorname{sech} y} = -\frac{1}{\sqrt{1-\operatorname{sech}^2 y}\,\operatorname{sech} y} \\
&= -\frac{1}{x\sqrt{1-x^2}},
\end{aligned}
$$

as required.

25)

Let $y = \operatorname{arcoth} x$, so $\coth y = x$. Then

$$
\begin{aligned}
\frac{d}{dx}(\coth y) &= \frac{d}{dx}(x), \\
-\operatorname{cosech}^2 y \frac{dy}{dx} &= 1, \\
\frac{dy}{dx} &= -\frac{1}{\operatorname{cosech}^2 y} = -\frac{1}{\coth^2 y - 1} = -\frac{1}{x^2-1} \\
&= \frac{1}{1-x^2},
\end{aligned}
$$

as required.

26 a)

$$
\begin{aligned}
y &= x^2 \operatorname{arcosech} 2x, \\
\frac{dy}{dx} &= x^2 \times 2 \times \left(-\frac{1}{(2x)\sqrt{(2x)^2+1}}\right) + 2x \operatorname{arcosech} 2x \\
&= -\frac{2x^2}{2x\sqrt{4x^2+1}} + 2x \operatorname{arcosech} 2x \\
&= -\frac{x}{\sqrt{4x^2+1}} + 2x \operatorname{arcosech} 2x.
\end{aligned}
$$

26 b)

$$
\begin{aligned}
\left.\frac{dy}{dx}\right|_{x=1/2} &= -\frac{1/2}{\sqrt{4(1/2)^2+1}} + 2\left(\frac{1}{2}\right) \operatorname{arcosech}\left(2 \times \frac{1}{2}\right) \\
&= -\frac{\sqrt{2}}{4} + \operatorname{arcosech} 1 \\
&= -\frac{\sqrt{2}}{4} + \ln(1+\sqrt{2}).
\end{aligned}
$$

27)

$$
\begin{aligned}
y &= \operatorname{arcsech} x^2 + \operatorname{arcoth} 3x, \\
\frac{dy}{dx} &= 2x \times \left(-\frac{1}{x^2\sqrt{1-(x^2)^2}}\right) + \frac{3}{1-(3x)^2}
\end{aligned}
$$

$$= -\frac{2}{x\sqrt{1-x^4}} + \frac{3}{1-9x^2}.$$

At $x = -\frac{1}{2}$,

$$\left.\frac{dy}{dx}\right|_{x=-1/2} = -\frac{2}{(-1/2)\sqrt{1-(-1/2)^4}} + \frac{3}{1-9(-1/2)^2}$$
$$= \frac{4}{\sqrt{15/16}} - \frac{12}{5} = \frac{16\sqrt{15}}{15} - \frac{12}{5}.$$